

UNIT 5

SQUARE ROOT OF POSITIVE NUMBER



PERFECT SQUARE

A perfect square is a number that is the result of multiplying a whole number by itself.

Examples of Perfect Squares:

- (i) $4 = 2 \times 2$ (2 squared)
- (ii) $9 = 3 \times 3$ (3 squared)
- (iii) $16 = 4 \times 4$ (4 squared)
- (iv) $25 = 5 \times 5$ (5 squared)
- (v) $36 = 6 \times 6$ (6 squared)

Perfect squares chart

X	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

Activity: Complete the table.

Number	Square	Perfect Square	Read As
1	$1^2 = 1 \times 1$	1	1 is the square of 1
2	$2^2 = \underline{\hspace{2cm}}$	4	4 is the square of 2
3	$(-3)^2 = (-3) \times (-3)$	9	<u> </u> is the square of <u> </u>
4	$5^2 = \underline{\hspace{2cm}}$	25	25 is the square of <u> </u>

Example:

Find whether the following number is a perfect squares or not.

(i) 5625

5	5625
5	1125
5	225
5	45
3	9
3	3
	1

Solution:

Prime factorization = $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$

Since each prime factor is in pairs,

So, 5625 is a perfect square.

Example:

Find the square of $\frac{3}{4}$ and compare with itself.

Solution:

Let us find square of $\frac{3}{4}$.

$$\left(\frac{3}{4}\right)^2 = \left(\frac{3}{4}\right) \times \left(\frac{3}{4}\right) = \frac{3 \times 3}{4 \times 4} = \frac{9}{16}$$

Example:

Find the square of 0.2 and compare with itself.

Solution:

$$\text{Square: } (0.2)^2 = (0.2) \times (0.2) = \frac{2}{10} \times \frac{2}{10} = \frac{4}{100} = 0.04$$

It is observed that: $0.04 < 0.2$
or $(0.2)^2 < 0.2$

Therefore, the square of a decimal less than one is always smaller than the given decimal.

Find the square root of 1024

$$1024 = 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2$$

$$\sqrt{1024} = 2 * 2 * 2 * 2 * 2$$

$$= 32 \text{ Ans}$$

2	1024
2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

EXERCISE 5.1

Q1. Using the prime factorization method, find which of the following numbers are perfect squares:

(i) 225 (ii) 189 (iii) 441 (iv) 729 (v) 1575 (vi) 900

Q2. Which of the following are the squares of even numbers?

(i) 196 (ii) 441 (iii) 400 (iv) 324 (v) 625

Q3. Find the squares of the following proper fractions and compare with themselves.

(i) $\frac{1}{5}$ (ii) $\frac{3}{7}$ (iii) $\frac{8}{9}$ (iv) $\frac{5}{6}$ (v) $\frac{2}{3}$

Q4. Find the squares of the following decimal numbers and compare with themselves.

(1) 0.1 (2) 0.5 (3) 0.07 (4) 0.11 (5) 0.15

Q5. Evaluate:

(i) $(38)^2 - (37)^2$ (ii) $(75)^2 - (74)^2$ (iii) $(92)^2 - (91)^2$
 (iv) $(105)^2 - (104)^2$ (v) $(141)^2 - (140)^2$ (vi) $(218)^2 - (217)^2$

SQUARE ROOT

The Square root of a number is a value that, when multiplied by itself, gives the original number. The symbol for square root is ' $\sqrt{\quad}$ '. Let us say m is a positive integer, such that $\sqrt{m \cdot m} = \sqrt{m^2} = m$. The square root is an inverse method of squaring a number. Suppose x is the square root of y , then it is represented as $x = \sqrt{y}$. For example, the square of 3 is 9, $3^2 = 9$ and the square root of 9, $\sqrt{9} = 3$. Since 9 is a perfect square, hence it is easy to find the square root.

Properties of Square root

Some of the important properties of the square root are as follows:

- If a number is a perfect square number, then there exists a perfect square root.
- If a number ends with an even number of zeros (0's), then it can have a square root.
- The two square root values can be multiplied. For example, $\sqrt{3}$ can be multiplied by $\sqrt{2}$, then the result should be $\sqrt{6}$.
- The square root of any negative number is not defined because the perfect square cannot be negative.
- If a number ends with 2, 3, 7 or 8 (in the unit digit), then the perfect square root does not exist.
- If a number ends with 1, 4, 5, 6 or 9 in the unit digit, then the number may have a perfect square root.

Find Square Root of Numbers

The methods to find the square root of numbers are:

- Square Root by Prime Factorization
- Square Root by Repeated Subtraction Method
- Square Root by Long Division Method
- Square Root by Estimation Method

Square root By Prime Factorization

The square root of a perfect square number is easy to calculate using the prime factorization method.

Let's solve some examples to understand this better.

Number	Prime Factorization	Square Root
16	$2 \times 2 \times 2 \times 2$	$\sqrt{16} = 2 \times 2 = 4$
144	$2 \times 2 \times 2 \times 2 \times 3 \times 3$	$\sqrt{144} = 2 \times 2 \times 3 = 12$
169	13×13	$\sqrt{169} = 13$
256	$256 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$	$\sqrt{256} = (2 \times 2 \times 2 \times 2) = 16$
576	$576 = 2 \times 2 \times 2 \times 2 \times 3 \times 3$	$\sqrt{576} = 2 \times 2 \times 2 \times 3 = 24$

Example:

Fine the square root of 784 by factorization method.

Solution:

Prime factorization of 784:

$$784 = \underline{2 \times 2 \times 2 \times 2} \times \underline{7 \times 7}$$

$$\sqrt{784} = 2 \times 2 \times 7 = 28$$

Answer = 28

2	784
2	392
2	196
2	98
7	49
7	7
	1

Example:

Fine the square root of 3969 by factorization method.

Solution:

Prime factorization of 3969:

$$3969 = \underline{3 \times 3 \times 3 \times 3} \times \underline{7 \times 7}$$

$$\sqrt{3969} = 3 \times 3 \times 7 = 63$$

Verification: $63 \times 63 = 3969$

Answer = 63

3	3969
3	1323
3	441
3	147
7	49
7	7
	1

Example:

Square Root of a Fraction

Fraction: $\frac{64}{81}$

Prime factorization of the numerator and denominator:

Numerator: $64 = 2 \times 2 \times 2 \times 2 \times 2 \times 2$

Denominator: $81 = 3 \times 3 \times 3 \times 3$

$$\begin{aligned}\sqrt{\frac{64}{81}} &= \frac{\sqrt{64}}{\sqrt{81}} = \frac{\sqrt{2 \times 2 \times 2 \times 2 \times 2 \times 2}}{\sqrt{3 \times 3 \times 3 \times 3}} \\ &= \frac{2 \times 2 \times 2}{3 \times 3} = \frac{8}{9}\end{aligned}$$

Answer: $\frac{8}{9}$

Numerator		Denominator	
2	64	3	81
2	32	3	27
2	16	3	9
2	8	3	3
2	4		1
2	2		
	1		

Example:

Square Root of a Decimal

Decimal: 1.44

Convert to fraction: $1.44 = \frac{144}{100}$

Prime factorization of the numerator and denominator:

Numerator: $144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3$

Denominator: $100 = 2 \times 2 \times 5 \times 5$

$$\begin{aligned}\sqrt{1.44} &= \sqrt{\frac{144}{100}} = \frac{\sqrt{144}}{\sqrt{100}} = \frac{\sqrt{2 \times 2 \times 2 \times 2 \times 3 \times 3}}{\sqrt{2 \times 2 \times 5 \times 5}} \\ &= \frac{2 \times 2 \times 3}{2 \times 5} = \frac{12}{10}\end{aligned}$$

Answer: 1.2

Numerator		Denominator	
2	144	2	100
2	72	2	50
2	36	5	25
2	18	5	5
3	9		1
3	3		
	1		

Square Roots by Long Division Method

Finding the square root of numbers that are not perfect squares can be a bit tricky, but we can easily calculate it using the long division method. Let's understand this step-by-step with an example.

$$\begin{array}{r}
 \begin{array}{r} 13 \\ \hline 1 \overline{) 180} \\ + 1 \overline{) -1} \\ \hline 23 \overline{) 080} \\ + 3 \overline{) 69} \\ \hline 26 \overline{) 1100} \end{array}
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{r} 13.4 \\ \hline 1 \overline{) 180} \\ + 1 \overline{) -1} \\ \hline 23 \overline{) 080} \\ + 3 \overline{) -169} \\ \hline 264 \overline{) 1100} \\ \overline{) 1056} \\ \hline 44 \end{array}
 \end{array}$$

Thus, the square root of 180 is 13.4.

Example:

Fine the square root of 7 by long division method.

$$\begin{array}{r}
 \begin{array}{r} 2.645 \\ \hline 2 \overline{) 7.0000} \\ + 2 \overline{) -4} \\ \hline 46 \overline{) 300} \\ + 6 \overline{) -276} \\ \hline 524 \overline{) 2400} \\ + 4 \overline{) -2096} \\ \hline 5285 \overline{) 30400} \\ + 5 \overline{) -26425} \\ \hline 5290 \overline{) 3975} \end{array}
 \end{array}$$

$$\sqrt{7} = 2.645$$

Example:

Square Root of $1\frac{1089}{1936}$ by Division Method.

Solution:

$$\text{As } 1\frac{1089}{1936} = \frac{3025}{1936}$$

Numerator:

$$\begin{array}{r} \overline{30\ 25} \\ 5\ 5 \\ \hline 5\ \overline{30\ 25} \\ + 5\ -\ 25 \\ \hline 10\ 5\ \overline{5\ 25} \\ + 5\ -\ 5\ 25 \\ \hline 110\ \overline{0} \end{array}$$

Denominator:

$$\begin{array}{r} \overline{19\ 36} \\ 4\ 4 \\ \hline 4\ \overline{19\ 36} \\ + 4\ -\ 16 \\ \hline 8\ 4\ \overline{3\ 36} \\ + 4\ -\ 3\ 36 \\ \hline 88\ \overline{0} \end{array}$$

$$\text{Hence, } \sqrt{1\frac{1089}{1936}} = \sqrt{\frac{3025}{1936}} = \frac{\sqrt{3025}}{\sqrt{1936}} = \frac{\cancel{55}^5}{\cancel{44}^4} = \frac{5}{4} = 1\frac{1}{4}$$

Example:

The length of a rectangular ground is twice its width. If its area is 96800 square metres, find the length of the rectangular ground.

Solution:

If the ground is divided into two equal parts, each will be a square.

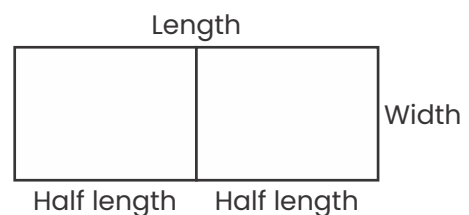
$$\text{The area of each square} = \frac{96800}{2}$$

$$= 48400 \text{ square meters.}$$

$$\begin{aligned} \text{Length of each square} &= \sqrt{48400} \\ &= 220 \text{ m.} \end{aligned}$$

Hence,

$$\begin{aligned} \text{length of the ground} &= 2 \times \text{length of square} \\ &= 2 \times 220 = 440 \text{ m.} \end{aligned}$$



$$\begin{array}{r} 2\ 20 \\ \hline 2\ \overline{4\ 84\ 00} \\ + 2\ -\ 4 \\ \hline 4\ 2\ \overline{8\ 4} \\ + 2\ -\ 8\ 4 \\ \hline 440\ \overline{000} \\ + 0\ -\ 000 \\ \hline 440\ \overline{0} \end{array}$$

EXERCISE 5.2

Q1. Find the square root of the following perfect squares:

(i) $\sqrt{16}$ (ii) $\sqrt{81}$ (iii) $\sqrt{144}$ (iv) $\sqrt{225}$

Q2. Find the Square Root by Factorization and Division Methods:

- | | | | |
|-----------------|-----------------|-----------------|-----------------|
| (1) 676 | (2) 169 | (3) 484 | (4) 961 |
| (5) 4900 | (6) 1089 | (7) 1600 | (8) 2304 |

WORD PROBLEMS

1. The area of a square field is 240.25 square metres. Calculate the length of its side.
2. The length of a room is three times its width, and the area is 720.75 square meters. Find its length and width.